



☑ : You are an expert in sports video analysis and trajectory-based motion prediction. I will provide you with a video of billiards...

✦ : <thinking> 1. Target Identification: The player is aiming the white cue ball at the black 8-ball. The target pocket appears to be the side pocket on the left side of the table (from the perspective of the camera after the shot is taken). 2. Motion Direction Analysis: The player strikes the cue ball, which then hits the 8-ball. The 8-ball travels on a straight trajectory directly towards the intended side pocket. The speed is moderate, suitable for a clean pocketing. 3. Outcome Prediction: The trajectory of the 8-ball is perfectly aligned with the center of the side pocket. There are no obstacles in its path, and its speed is appropriate. Based on its movement right up to the edge of the pocket before the video ends, it is clear that the ball will fall in. </thinking> <answer> Yes </answer>

(a) Target Identification Error



☑ : You are an expert in sports video analysis and trajectory-based motion prediction. I will provide you with a video of a golf shot...

✦ : <thinking> 1. Initial Motion Analysis: The video begins with golfer Wade Ormsby taking a tee shot on a par 3 hole. We see his full swing and follow-through. While the ball's flight isn't clearly tracked against the sky, the golfer's reaction indicates a solid strike. 2. Trajectory Tracking: The camera cuts to the green to pick up the ball's landing and roll. At 00:08, the ball comes into view, bouncing and rolling on the green. It's moving from left to right towards the pin. 3. Outcome Prediction: Watching the final segment of the roll, the ball approaches the hole but continues past it to the left (from the camera's perspective). The video ends with the ball having clearly missed the hole and come to rest past it. Therefore, the ball does not go into the hole. </thinking> <answer> No </answer>

(b) Target Tracking Error



☑ : You are an expert in driving video analysis and subtle motion detection. I will provide you with a driving video...

✦ : <thinking> 1. Target Identification: The vehicle in the green box is a dark car driving in the right lane, relatively close to the ego vehicle. The vehicle in the blue box is a light-colored SUV driving in the left lane, further ahead of the green car. 2. Motion Analysis: By observing the video from start to finish, we can see the relative distance between the two vehicles. Initially, the distance between the green car and the blue SUV is relatively small. As the video progresses, the blue SUV noticeably pulls away from the green car, increasing the gap between them. 3. Speed Comparison: Because the distance between the two vehicles is increasing, with the blue vehicle moving further ahead of the green vehicle over time, it is evident that the vehicle in the blue box is traveling at a higher speed than the vehicle in the green box. </thinking> <answer> Blue </answer>

(c) Motion State Error



☑ : You are an expert in sports video analysis and physics trajectory prediction. I will provide you with a video of a basketball shot. Your task is to predict whether the basketball will go into the hoop...

✦ : <thinking> 1. Release Analysis: The player in the white shirt shoots the ball from close range. The release seems slightly rushed due to the presence of defenders. The direction of the shot appears to be slightly off-center, aiming more towards the right side of the rim rather than the direct center. 2. Trajectory Tracking: As the ball travels through the air, it heads towards the right side of the basketball hoop. It makes contact with the right edge of the rim and then bounces off the backboard. 3. Outcome Prediction: Based on the ball hitting the right side of the rim and bouncing off the backboard away from the center of the hoop, it is clear that the shot is not successful. The trajectory was not aligned correctly to pass through the net. </thinking> <answer> No </answer>

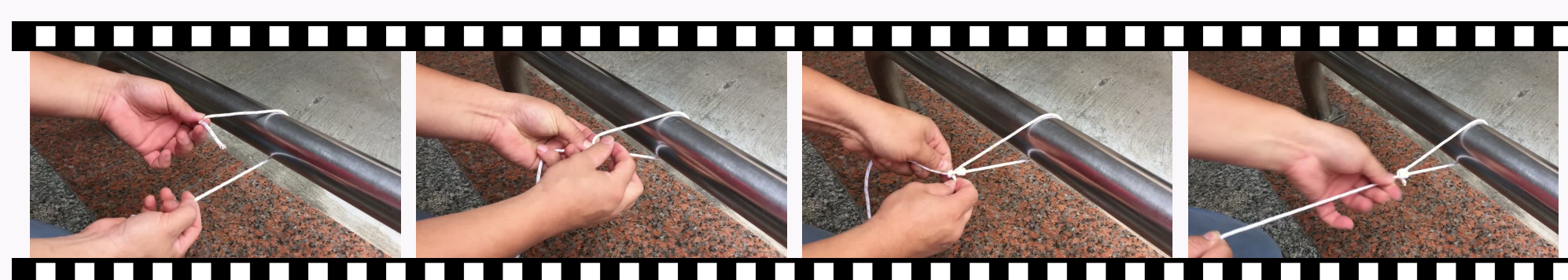
(d) Outcome Reasoning Error



☑ : You are an expert in driving scene analysis and vehicle-obstacle clearance reasoning. I will provide you with a video showing a vehicle moving near traffic cones. Your task is to predict whether the vehicle can complete the current driving maneuver without touching any cone...

✦ : <thinking> 1. Cone Layout Analysis: There are two traffic cones placed on the ground, one on the left and one on the right, forming a clearly defined narrow passage for vehicles to pass through. 2. Vehicle Motion Analysis: A grey SUV is moving straight forward, approaching the passage formed by the two cones. The vehicle is well-aligned with the center of the gap between the cones and is moving at a slow, steady pace without any significant steering deviations. 3. Clearance Prediction: As the vehicle enters the space between the cones, it remains centered. There is visible clearance between the vehicle's sides and both the left and right cones. The vehicle's straight trajectory and centered position indicate it will successfully pass through the narrow passage without touching either cone. </thinking> <answer> Yes </answer>

(e) Spatial Relation Error



☑ : You are an expert in physical reasoning and knot behavior analysis. I will provide you with a video of knot manipulation. Your task is to determine whether the knot is a slip knot or a fixed knot...

✦ : <thinking> 1. Structure Observation: In the video, a string is passed around a metal rail. The person then ties a knot by forming a bight (a folded loop) with one end of the string and passing it through a loop made by the other end. The resulting knot has a distinct loop sticking out and a free tail end hanging from it. This indicates a slipped structure. 2. Pulling Behavior Analysis: Towards the end of the video, the person grabs the free tail end of the string that is connected to the protruding loop. When they pull this single tail end, the loop is drawn back through the knot's structure. As soon as the loop pulls through completely, the entire knot loses its binding friction and instantly falls apart. 3. Knot Type Determination: Since pulling one specific end of the rope (the tail of the slipped loop) causes the knot to completely release and unravel, it perfectly matches the definition of a slip knot. </thinking> <answer> Slip </answer>

(f) Physical Interaction Error